CSE370:Database Systems

# Lecture 6: Normalization For Relational Databases



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Inspiring Excellence

# Database Normalization

Database normalization is the process of decomposing (breaking down) inefficient relations (tables) into smaller relations in order to minimize data redundancy and avoid update anomalies.

# Data Redundancy Causes

- Storage Wastage
- Update Anomalies
  - →Insertion Anomaly
  - →Deletion Anomaly
  - →Modification Anomaly

# INSERTION ANOMALY

Some data cannot be inserted at all, even though it should be inserted according to business rules.

A new employee cannot be inserted in the EMP_PROJ table unless a project is assigned to the employee, even though it is not the expected insertion rule of the company

Redundancy | Redundancy

EMP_PROJ

| Ssn | Pnumber | Hours | Ename | Pname | Plocation |
|---|---|---|---|---|---|
| 123456789 | 1 | 32.5 | Smith, John B. | ProductX | Bellaire |
| 123456789 | 2 | 7.5 | Smith, John B. | ProductY | Sugarland |
| 666884444 | 3 | 40.0 | Narayan, Ramesh K. | ProductZ | Houston |
| 453453453 | 1 | 20.0 | English, Joyce A. | ProductX | Bellaire |
| 453453453 | 2 | 20.0 | English, Joyce A. | ProductY | Sugarland |
| 333445555 | 2 | 10.0 | Wong, Franklin T. | ProductY | Sugarland |
| 333445555 | 3 | 10.0 | Wong, Franklin T. | ProductZ | Houston |
| 333445555 | 10 | 10.0 | Wong, Franklin T. | Computerization | Stafford |
| 333445555 | 20 | 10.0 | Wong, Franklin T. | Reorganization | Houston |
| 999887777 | 30 | 30.0 | Zelaya, Alicia J. | Newbenefits | Stafford |
| 999887777 | 10 | 10.0 | Zelaya, Alicia J. | Computerization | Stafford |
| 987987987 | 10 | 35.0 | Jabbar, Ahmad V. | Computerization | Stafford |
| 987987987 | 30 | 5.0 | Jabbar, Ahmad V. | Newbenefits | Stafford |
| 987654321 | 30 | 20.0 | Wallace, Jennifer S. | Newbenefits | Stafford |
| 987654321 | 20 | 15.0 | Wallace, Jennifer S. | Reorganization | Houston |
| 888665555 | 20 | Null | Borg, James E. | Reorganization | Houston |

# DELETION ANOMALY

Some data is unintentionally lost due to the deletion of some other data.

The employee named “Narayan” will also be deleted if the project with pnumber 3 is deleted since that employee is assigned only to one project, even though the intention was only to delete the project.

Redundancy | Redundancy

EMP_PROJ

| Ssn | Pnumber | Hours | Ename | Pname | Plocation |
|---|---|---|---|---|---|
| 123456789 | 1 | 32.5 | Smith, John B. | ProductX | Bellaire |
| 123456789 | 2 | 7.5 | Smith, John B. | ProductY | Sugarland |
| 666884444 | 3 | 40.0 | Narayan, Ramesh K. | ProductZ | Houston |
| 453453453 | 1 | 20.0 | English, Joyce A. | ProductX | Bellaire |
| 453453453 | 2 | 20.0 | English, Joyce A. | ProductY | Sugarland |
| 333445555 | 2 | 10.0 | Wong, Franklin T. | ProductY | Sugarland |
| 333445555 | 3 | 10.0 | Wong, Franklin T. | ProductZ | Houston |
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# MODIFICATION ANOMALY

Same data is stored on multiple rows redundantly, incorrect update may result in data inconsistencies.

Plocation of “ProductX” is updated to “Sugarland” from “Bellaire” for the first row, SSN=’123456789' and Pnumber=1. However, in other rows PLocation for “ProductX” remains “Bellaire”. Thus the data becomes inconsistent.

Redundancy Redundancy

EMP_PROJ

| Ssn | Pnumber | Hours | Ename | Pname | Plocation |
|---|---|---|---|---|---|
| 123456789 | 1 | 32.5 | Smith, John B. | ProductX | Bellaire |
| 123456789 | 2 | 7.5 | Smith, John B. | ProductY | Sugarland |
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| 333445555 | 2 | 10.0 | Wong, Franklin T. | ProductY | Sugarland |
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| 987987987 | 10 | 35.0 | Jabbar, Ahmad V. | Computerization | Stafford |
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# Functional Dependencies (FDs)

**FDs are constraints that are derived from the meaning and interrelationships of the attributes of a table.**

| X | Y |
|---|---|
| 1 | a |
| 2 | b |
| 1 | a |
| 3 | b |

- Formally FDs are shown using the notation X → Y
- It means, a set of attributes X functionally determines a set of attributes Y (i.e. Y is functionally dependent on X) such that whenever two rows have the same value for X, the corresponding value for Y must also be the same.
  - If all values of X are unique, then X → Y always holds. If X values are repeated, then X → Y is only true if for each repeated value of X, the corresponding value of Y is the same.

# Examples of FDs

- **ID → {Name, CGPA}**
  The student's ID determines the students's name and cgpa in the Student table

- **{ID, CCode} → Grade**
  The Grade is dependent on the ID of the Student and the course code in the 'Grades' table

  - FDs are derived from real-world constraints on the attributes. The primary key of a table will functionally determine all other attributes of the table.

  - A non-prime attribute (attribute not part of pk) may also be functionally dependent on other non-prime attributes.

Student

| ID | Name | CGPA |
|---|---|---|
| 1 | Luffy | 4.0 |
| 2 | Sanji | 3.5 |
| 3 | Zoro | 3.8 |

Grades

| ID | Ccode | Grade |
|---|---|---|
| 1 | CSE370 | A |
| 2 | CSE370 | B+ |

# Normal Forms

**Normal form of a relation(table) indicates the degree to which the relation has been normalized.**

- The normal forms are **1NF(first normal form)**, **2NF(second normal form)**, **3NF(third normal form)**, **BCNF(Boyce-Codd normal form)**, **4NF(fourth normal form)** and **5NF(fifth normal form)**.

- Each normal form has some conditions using the keys and FDs of a relation(table). These conditions are used to certify whether that relation is in a particular normal form.

- In practice, schemas are typically normalized up to 3NF (and sometimes BCNF). Higher normal forms, will increase the number of tables, thus will require more joins and negatively impact query performance. Therefore, over-normalization is not practiced in real database systems.

# First Normal Form (1NF)

1NF is considered to be part of the definition of a relation

## DISALLOWS

- Composite Attributes
- Multivalued Attributes
- Nested Relations (tables); attributes whose values for an individual tuple are non-atomic

# Normalize to 1NF

The table below is not in 1NF form as DLocations is multivalued

Department

| DName | Dnumber | DmgrID | DLocations |
|---|---|---|---|
| Research | 5 | E1 | {Dhaka, Sylhet, Khulna} |
| Admin | 4 | E2 | {Dhaka} |
| HQ | 1 | E3 | {Chittagong} |

Department

| DName | Dnumber | DmgrID |
|---|---|---|

Dept_Loc

| Dnumber | DLocations |
|---|---|

A separate table has been created for the multivalued attribute

# Second Normal Form (2NF)

**Prime attribute**: an attribute which is part of the pk
**Non-prime attribute**: an attribute which is not part of the primary key

## TO BE IN 2NF, A RELATION:

- Must also be in 1NF
- Cannot have any partial functional dependency on primary key, thus all non-prime attributes must be fully functionally dependent on the primary key

| StudentID | CourseCode | StudentName | Grade |
|---|---|---|---|

- The primary key of the above table is {StudentID, CourseCode}
- {StudentID, CCode} → Grade. Since Grade depends on the full primary key, it is a full functional dependency
- StudentID → StudentName. Student's name depends only on their ID. Since, a non-prime attribute depends on a part of the primary key, it is a partial functional dependency

# Normalize to 2NF

The table below is not in 2NF form as partial dependency exists on primary key due to FD2 and FD3.

Emp_Proj

| EID | Pno | Hours | Ename | Pname | Plocation |
|---|---|---|---|---|---|

Functional Dependencies:

FD1: EID, Pno → Hours
FD2: EID → Ename
FD3: Pno → Pname, Plocation

Employee

| EID | Ename |
|---|---|

Project

| Pno | Pname | Plocation |
|---|---|---|

Emp_Proj

| EID | Pno | Hours |
|---|---|---|

Note: If functional dependency of any non-prime attribute is not mentioned, then assume full dependency on primary key.

# Third Normal Form (3NF)

If X → Y and Y → Z, where X is pk and Y is non-prime attribute, then X → Z is a transitive dependency

## TO BE IN 3NF, A RELATION:

- Must also be in 2NF
- Cannot have any transitive functional dependency, thus no non-prime attribute should be transitively dependent on the primary key

| <u>StudentID</u> | StudentName | DeptID | DName |
|---|---|---|---|

- The primary key of the above table is StudentID
- StudentID → DeptID and DeptID → DName, therefore, StudentID → DName is a transitive dependency
- StudentID → StudentName is a non-transitive dependency

# Normalize to 3NF

The table below is not in 3NF form as transitive dependency exists on primary key due to FD2.

Emp_Dept

| EID | Ename | DOB | Address | Dno | Dname | DMgr |
|---|---|---|---|---|---|---|

Functional Dependencies:

FD1: EID → Ename, DOB, Address, Dno
FD2: DNo → DName, DMgr

Employee

| EID | Ename | DOB | Address | Dno |
|---|---|---|---|---|

Department

| Dno | Dname | DMgr |
|---|---|---|

Note: If any non-prime attribute is functionally dependent on any other non-prime attribute, it creates transitive dependency on the primary key

# Normalization Example

Student

| StID | Course Code | Semester | Student Name | Attempted Credits | Course Title | Grade | Proj ID | Project Marks | Project Name |
|---|---|---|---|---|---|---|---|---|---|

**Additional Functional Dependencies**

FD1: StudentID → StudentName
FD2: CourseCode → CourseTitle
FD3: StudentID, Semester → AttemptedCredits
FD4: ProjID → ProjectName, ProjectMarks

**Question:** Identify and explain which normal form the above schema is in. Step by step normalize it until 3rd normal form. You MUST show the complete “current” schema after each normalization step.

# Normalization Example - Solution (1)

The given schema is in first normal form (1NF) because there are no multivalued/composite attributes and there are no nested tables. It is not in second normal form (2NF) because partial functional dependency on primary key exists due to FD1, FD2 and FD3.

**2NF Normalization**

Student

| StID | StudentName |
|---|---|

Course

| CourseCode | Course Title |
|---|---|

Student_Credits

| StID | Semester | Attempted Credits |
|---|---|---|

Student_Grades

| StID | CourseCode | Semester | Grade | ProjID | Project Marks | Project Name |
|---|---|---|---|---|---|---|

# Normalization Example - Solution (2)

The current schema is in 2NF but not in third normal form (3NF) yet, because transitive functional dependency on primary key exists due to FD4 in Student_Grades table.

**3NF Normalization**

Student

| StID | StudentName |
|---|---|

Course

| CourseCode | Course Title |
|---|---|

Student_Credits

| StID | Semester | Attempted Credits |
|---|---|---|

Student_Grades

| StID | Course Code | Semester | Grade | ProjID |
|---|---|---|---|---|

Student_Projects

| ProjID | Project Marks | Project Name |
|---|---|---|

# What Next?

## LECTURE 7: NORMALIZATION



LOADING.....

